

Spot Buy Limit Order — Complete Test Steps

File: `tests/trade/spotBuyLimitOrder.spec.ts`

Data: `src/data/spotBuyLimitData.csv`

Mode: Serial (each step depends on the previous one)

SETUP — Before All Tests

Step 1: Launch a Chromium browser (non-headless, 1280×720 viewport) and open a new browser page.

Step 2: Navigate to the login page (`BASE_URL` + login route).

Step 3: Enter the email address in the email field and click "Continue".

Step 4: Enter the password in the password field and click "Continue".

Step 5: Read the current TOTP/OTP value and enter it in the OTP field, then click "Continue" to complete login successfully.

Step 6: Dismiss any post-login dialogs (popups, banners, announcements) and wait for the home/dashboard page to be fully loaded.

TC-01 — Navigate to Spot Trading

Step 7: Click on the Trading menu item and navigate to the Spot Trading page; wait for the page to fully load.

TC-02 — Verify Page Labels

Step 8: Read all visible labels on the Spot Trading page and verify each one is correct:

- "Trading" page heading is present
- "Spot" label is present
- "Depth View" label is present
- "Order Book" heading is present
- "Buy" tab is present
- "Sell" tab is present
- "Limit" tab is present
- "Market" tab is present

- "Stop" tab is present
 - "Market Trades" label is present
 - "My Trades" label is present
 - "Open Orders" tab is present
 - "All Orders" tab is present
 - "Trade History" label is present
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TC-03 — Search Currency Pair

Step 9: Type `BTC/USDT` (read from CSV `searchPair` column) into the market search input and wait for results to appear.

TC-04 — Mark as Favorite

Step 10: Click the star icon next to `BTC/USDT` in the search results to mark it as a favorite.

Step 11: Verify the success toast message confirms the pair was **added** to favorites.

TC-05 — Unmark from Favorites

Step 12: Click the star icon again next to `BTC/USDT` to unmark it from favorites.

Step 13: Verify the status message confirms the pair was **removed** from favorites, and "No records found" is visible in the Favorites tab.

TC-06 — Select Currency Pair

Step 14: Click the "ALL" tab in the pair list, find `BTC/USDT`, and click it to activate it as the trading pair. The whole page (order book, ticker, trade form) now shows data for this pair.

TC-07 — Verify 24h Ticker vs Binance

Step 15: Simultaneously fetch the 24h ticker data from the page and from the Binance public API for `BTC/USDT`, then verify:

- Page last price is a positive number
- Page last price is within 1% of Binance last price (live price tolerance)
- Page 24h high matches Binance exactly
- Page 24h low matches Binance exactly

- Page 24h base volume is within 1% of Binance base volume
 - Page 24h quote volume difference is logged as informational only (exchange counts both sides of each trade, approximately 2× Binance — not a bug)
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TC-08 — Order Book Column Headers

Step 16: Read the order book column headers and verify all three are visible and non-empty: **Price, Amount, Total**.

TC-09 — Order Book View Switches

Step 17: Switch the order book to "Sell-only" view; verify the order book is still visible.

Step 18: Switch to "Buy-only" view; verify the order book is still visible.

Step 19: Switch back to "All" view (restoring the default).

TC-10 — Order Book Precision Dropdown

Step 20: Cycle through precision values `0.01` → `0.1` → `1` → `0.01` (restoring the default). For each precision, and for each of the three views (all / sell / buy), verify that displayed prices have no more decimal places than the precision allows:

- `0.01` → max 2 decimal places
- `0.1` → max 1 decimal place
- `1` → max 0 decimal places

Restore to "All" view at `0.01` precision at the end.

TC-11 — Order Book LTP and Buy/Sell Ratio Bar (Suggestions)

Step 21: Attempt to read the Last Traded Price (LTP) from the order book mid-row. Record as a **suggestion** that LTP display is not yet implemented. No assertion failure.

Step 22: Attempt to read the buy % and sell % ratio bar values. Record as a **suggestion** that the ratio bar is not yet implemented. No assertion failure.

TC-12 — Order Book Has Real Data Rows

Step 23: Count the ask rows and bid rows in the order book, read the top ask price and top bid price. Verify:

- Ask row count > 0
 - Bid row count > 0
 - Top ask price > 0
 - Top bid price > 0
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TC-13 — Valid Spread

Step 24: Read the top bid and top ask prices. Verify that top bid < top ask (a valid market spread exists).

TC-14 — Bid/Ask vs Binance within 0.5%

Step 25: Simultaneously fetch the top bid/ask from the page and from the Binance order book API. Verify:

- Page top bid is within 0.5% of Binance top bid
 - Page top ask is within 0.5% of Binance top ask
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TC-15 — Order Book LTP vs Binance within 0.5%

Step 26: Simultaneously read the page order book LTP and the Binance 24h last price. Verify the LTP is positive and within 0.5% of the Binance last price.

TC-16 — Precision Toggle Restores LTP

Step 27: Read the current order book precision. Switch to an alternate precision (e.g., 0.1 if current is 0.01, or vice versa). Wait briefly, then restore the original precision. Verify the order book LTP is still a positive number after the toggle.

TC-17 — Order Book Ask Rows Pre-Fill Buy Form

Step 28: Click the Buy tab, then click the Limit tab to activate the Buy Limit form.

Step 29: Click each of the top 5 ask rows in the order book one at a time. After each click, verify that the Price input field shows the row's price and the Amount input field is filled. Verify all 5 rows pass the price and amount fill check.

TC-18 — Buy Button Label

Step 30: Read the text on the Buy button. Verify it contains "BUY" and the base coin name (e.g., "BUY BTC").

TC-19 — Sell Tab Available Balance

Step 31: Switch to the Sell tab. Read the "Available" balance shown for the base coin (BTC). Verify it is ≥ 0 .

TC-20 — 25% Button

Step 32: Switch back to the Buy tab and click the Limit tab. Enter the limit price of `57000` in the price field. Click the **25%** button. Verify the total field is filled with a positive value close to 25% of the available USDT balance (within 5% tolerance).

TC-21 — 50% Button

Step 33: Click the **50%** button (limit price `57000` already set). Verify the total field is filled close to 50% of the available USDT balance (within 5% tolerance).

TC-22 — 75% Button

Step 34: Click the **75%** button. Verify the total field is filled close to 75% of the available USDT balance (within 5% tolerance).

TC-23 — 100% Button

Step 35: Click the **100%** button. Verify the total field is filled close to 100% of the available USDT balance (within 5% tolerance).

TC-24 — Pair Header Shows Selected Pair

Step 36: Read the pair header/title shown above the order form. Verify it contains "BTC" or "BTC/USDT".

TC-25 — Capture Pre-Order Balance Snapshot

Step 37: Navigate to the **Portfolio** → **Spot** page and capture a full balance snapshot including:

- Available USDT balance (buy side)
- Available BTC balance (sell side)
- Portfolio spot balances for both USDT and BTC
- Funds balances for both USDT and BTC

Navigate back to the Spot Trading page. Verify the buy available balance (USDT) is ≥ 0 and matches the portfolio USDT spot balance.

TC-26 — Enter Limit Order Details and Verify Estimated Fee

Step 38: Click the Buy tab and Limit tab to activate the Buy Limit form. Enter the limit price `57000` USDT and total `8` USDT (values from CSV `buyLimitPrice` and `total` columns).

Step 39: Read the estimated fee shown on the form. Calculate the expected fee as:

`total / price × price × feePercent ÷ 100` (feePercent `0.15%` from CSV).

Verify the UI estimated fee matches the calculated fee.

TC-27 — Total Field Auto-Calculates

Step 40: Read the current price and amount fields. Verify the Total field value equals price × amount within 1% tolerance.

TC-28 — Fetch Available Balance Before Order

Step 41: Read and log the current USDT available balance (from the snapshot captured in TC-25). Verify it is ≥ 0 . This is the baseline for the post-order comparison.

TC-29 — Confirm Buy Order and Verify Success

Step 42: Check if the text "Insufficient balance" is visible on the form within 2 seconds.

- If **YES** → mark order as failed, record a soft failure with the message "Insufficient balance — top up and re-run", and stop this test. All dependent tests will be skipped.

Step 43: If balance is sufficient → click the **BUY BTC** button to submit / open the confirmation dialog.

Step 44: In the confirmation dialog, read the executed price, executed amount, and order ID. Click **Confirm** to finalize the order.

Step 45: Wait for the success message/toast. Verify the message contains "success", "created", "placed", or "status:".

Step 46: Fetch the current Binance market price to detect whether the limit price (57000) is above the market price — if so, the order fills immediately; otherwise it remains pending.

TC-30 — Balance Snapshot After Placing Order

Skipped if TC-29 failed.

Step 47: Navigate to Portfolio and capture a new full balance snapshot.

Step 48: Compare the before-snapshot (TC-25) vs the after-snapshot:

- If the order is **pending** (limit price below market): verify USDT available balance decreased by the order total (8 USDT), locked in inOrder.
 - If the order **filled immediately** (limit price above market): verify USDT decreased and BTC increased by the correct amounts within 0.5% tolerance.
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TC-31 — Market Trades Panel Headers

Step 49: Click the **Market Trades** tab in the bottom panel. Read the column headers. Verify they include "price", "amount" (or "qty"), and "time".

TC-32 — My Trades Panel Headers

Step 50: Click the **My Trades** tab. Read its column headers. Verify they include "price", "amount" (or "qty"), and "time". Switch back to the Market Trades tab.

TC-33 — Market Trades Has Real Data Rows

Step 51: Read the data rows in the Market Trades panel. Verify:

- At least 1 row exists
 - The first row's price is positive
 - The first row's amount is positive
 - The first row has a non-empty time string
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TC-34 — Market Trades Prices Match Binance within 1%

Step 52: Simultaneously read the Market Trades rows from the page and fetch the 20 most recent trades from the Binance API. Verify the most recent page trade price is within 1% of the Binance most recent trade price.

TC-35 — Open Orders Section Visible

Step 53: Read the Open Orders section. Verify the "Open Orders" tab text is present and the "View All" button is present.

TC-36 — Open Orders Shows the Placed Buy Limit Order

Skipped if TC-29 failed.

Step 54: If the order filled immediately (limit price was above market), log a suggestion that no pending entry appears in Open Orders and skip assertions.

Step 55: Otherwise, open the **Open Orders** tab and verify the latest order row shows:

- Pair contains "BTCUSDT"
 - Type contains "limit"
 - Side contains "Buy New"
 - Price > 0 (approximately 57000)
 - Amount > 0
 - Filled = 0
 - Remaining > 0
 - Stop Limit = "-"
 - Cancel button is present (confirming the order is pending and cancellable)
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TC-37 — All Orders Shows the Placed Buy Limit Order

Skipped if TC-29 failed.

Step 56: Open the **All Orders** tab and find the latest row. Verify:

- Pair contains "BTCUSDT"
 - Type contains "limit"
 - Side contains "Buy New"
 - Limit price field > 0
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- If **pending**: status matches `new / pending / open` , filled = 0, remaining > 0
 - If **immediately filled**: status matches `done / filled / complete` , filled > 0
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TC-38 — Available Balance Decreases After Order

Skipped if TC-29 failed.

Step 57: Read the All Orders total for the placed order. Verify it does not exceed the entered total (8 USDT) by more than 0.1%, and the percentage difference is less than 6%. Verify the current buy available balance (USDT) has decreased by the locked amount.

TC-39 — Trade History Shows the Placed Order (Skipped if Pending)

Skipped if TC-29 failed.

Step 58: Open the **Trade History** tab and search for the order by order ID.

- If **not found** → log a suggestion ("order is still pending — Trade History only shows executed trades") and skip assertions.

Step 59: If found, verify:

- Pair contains "BTCUSDT"
 - Side contains "buy"
 - Executed price > 0
 - Amount > 0
 - Total > 0
 - Fee ≥ 0
 - Date/time is within 60 seconds of when the order was placed
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TC-40 — Trade History Bottom Tab Shows the Placed Order

Skipped if TC-29 failed.

Step 60: Read the first row of the Trade History widget at the bottom of the trading page. If the first row's order ID does not match the placed order, log a suggestion (order still pending) and skip assertions.

Step 61: If the row matches, verify:

- Pair references "BTC" or "BTCUSDT"
- Side contains "buy"

- Price is positive

TC-41 — Cancel Latest Open Order

Skipped if TC-29 failed.

Step 62: If the order filled immediately, log a suggestion that there is nothing to cancel and skip.

Step 63: Otherwise, click **Cancel** on the first row in Open Orders; confirm the cancel in the modal; verify the success toast confirms cancellation and the USDT balance is restored.

TC-42 — Full Balance Snapshot After Cancel

Skipped if TC-29 failed.

Step 64: If the order filled immediately, skip.

Step 65: Navigate to Portfolio and capture a new balance snapshot. Verify:

- USDT available balance has been restored (inOrder returns to 0)
- BTC balance unchanged (no trade took place)
- All differences are within 0.5% tolerance

TC-43 — Snapshot Before Multi-Order Cancel All Test

Skipped if TC-29 failed.

Step 66: Navigate to Portfolio and capture a full balance snapshot before placing multiple pending orders. Verify both USDT available balance and BTC available balance are ≥ 0 . This snapshot is the baseline for the Cancel All balance restoration check.

TC-44 — Place 1st Pending Buy Limit Order

Skipped if TC-29 failed.

Step 67: Enter limit price `55000` USDT and total `10` USDT (from CSV `buyLimitPrice2 / buyLimitTotal2`). Click BUY BTC and confirm. Verify success message contains "success".

Step 68: Verify in All Orders that the order status is `new / pending / open` . Verify Open Orders shows ≥ 1 pending row. Verify the order does NOT appear in Trade History. Verify USDT available balance decreased by the locked amount.

TC-45 — Place 2nd Pending Buy Limit Order

Skipped if TC-44 failed.

Step 69: Enter limit price `54000` USDT and total `8` USDT (from CSV `buyLimitPrice3 / buyLimitTotal3`). Click BUY BTC and confirm. Verify success message contains "success".

Step 70: Verify All Orders status is `new / pending / open` . Verify Open Orders shows ≥ 2 rows. Verify order is NOT in Trade History. Verify USDT available balance decreased by the locked amount.

TC-46 — Place 3rd Pending Buy Limit Order

Skipped if TC-44 failed.

Step 71: Enter limit price `53000` USDT and total `7` USDT (from CSV `buyLimitPrice4 / buyLimitTotal4`). Click BUY BTC and confirm. Verify success message contains "success".

Step 72: Verify All Orders status is `new / pending / open` . Verify Open Orders shows ≥ 3 rows. Verify order is NOT in Trade History. Verify USDT available balance decreased by the locked amount.

TC-47 — Place 4th Pending Buy Limit Order

Skipped if TC-44 failed.

Step 73: Enter limit price `52000` USDT and total `9` USDT (from CSV `buyLimitPrice5 / buyLimitTotal5`). Click BUY BTC and confirm. Verify success message contains "success".

Step 74: Verify All Orders status is `new / pending / open` . Verify Open Orders shows ≥ 4 rows. Verify order is NOT in Trade History. Verify USDT available balance decreased by the locked amount.

TC-48 — Place 1st Pending Sell Limit Order (for Cancel All)

Skipped if TC-44 failed.

Step 75: Switch to the Sell tab and Limit tab. Enter limit price `72000` USDT and amount `0.00011` BTC (from CSV `cancelAllSellPrice1 / cancelAllSellAmount1`). Click SELL BTC and confirm. Verify success message contains "success".

Step 76: Verify All Orders status is `new / pending / open` . Verify Open Orders shows ≥ 5 rows. Verify order is NOT in Trade History. Verify BTC available balance decreased by the locked amount.

TC-49 — Place 2nd Pending Sell Limit Order (for Cancel All)

Skipped if TC-44 failed.

Step 77: Enter limit price `73000` USDT and amount `0.00012` BTC (from CSV `cancelAllSellPrice2 / cancelAllSellAmount2`). Click SELL BTC and confirm. Verify success message contains "success".

Step 78: Verify All Orders status is `new / pending / open` . Verify Open Orders shows ≥ 6 rows. Verify order is NOT in Trade History. Verify BTC available balance decreased by the locked amount.

TC-50 — Open Orders Shows All 6 Pending Orders

Skipped if TC-44 failed.

Step 79: Open the Open Orders tab and verify all 6 pending orders are shown (4 buy + 2 sell). Verify the **Cancel All** button is visible.

TC-51 — Balance Shows Buy and Sell Orders Locked in inOrder

Skipped if TC-44 failed.

Step 80: Navigate to Portfolio and compare the current balance against the pre-multi-order snapshot (TC-43). Verify:

- USDT inOrder has increased by the sum of all 4 buy order totals (approximately `10 + 8 + 7 + 9 = 34` USDT)
 - BTC inOrder has increased by the sum of all 2 sell order amounts (approximately `0.00011 + 0.00012 = 0.00023` BTC)
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TC-52 — Trade History Does NOT Show Pending Orders

Skipped if TC-44 failed.

Step 81: Click the **Trade History** tab. Log the number of rows visible. Verify (informational, no assertion failure) that the pending limit orders placed in TC-44 through TC-49 do not appear in Trade History — only executed trades appear there.

TC-53 — Cancel One Order Individually

Skipped if TC-44 failed.

Step 82: Click the **Open Orders** tab. Click the **Cancel** button on the first row in Open Orders. Confirm the cancel in the modal. Verify the success toast confirms the individual cancellation.

TC-54 — 5 Orders Remain After Individual Cancel

Skipped if TC-44 failed.

Step 83: Click the Open Orders tab again and count the remaining rows. Verify at least 5 rows remain (6 placed minus 1 cancelled).

TC-55 — Cancel All Removes All Remaining Orders

Skipped if TC-44 failed.

Step 84: Click the **Cancel All** button in the Open Orders section. Confirm in the modal. Verify the success toast contains "All order cancelled successfully".

TC-56 — Open Orders Shows "No data" After Cancel All

Skipped if TC-44 failed.

Step 85: Verify Open Orders shows "No data" text and the row count is 0.

TC-57 — All Orders Shows Cancelled Status

Skipped if TC-44 failed.

Step 86: Open the **All Orders** tab. Verify the most recent order row shows a cancelled/done status with red color text.

TC-58 — Trade History Does Not Show Cancelled Pending Orders

Skipped if TC-44 failed.

Step 87: Open the **Trade History** tab. Log the row count. Verify (informational, no assertion failure) that the cancelled pending orders do not appear here — cancelled unfilled orders are never recorded in Trade History.

TC-59 — Balance Fully Restored After Cancel All

Skipped if TC-44 failed.

Step 88: Navigate to Portfolio and capture a new balance snapshot. Compare against the pre-multi-order snapshot (TC-43). Verify:

- USDT available balance restored (inOrder returns to 0) within 0.5% tolerance
 - BTC available balance restored (inOrder returns to 0) within 0.5% tolerance
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TC-60 — Snapshot Before Above-Market Order

Skipped if TC-29 failed.

Step 89: Navigate to Portfolio and capture a full balance snapshot before placing the above-market buy. Verify USDT available balance is ≥ 0 .

TC-61 — Place Buy Limit Order Above Market Price

Skipped if TC-29 failed.

Step 90: Switch to the Buy Limit form. Enter limit price `70000` USDT and total `7` USDT (from CSV `aboveMarketLimitPrice / aboveMarketTotal`). Click BUY BTC and confirm.

Step 91: Verify success message contains "success" or "created". Verify the executed price is $\leq$ the limit price of `70000` (the order fills at the best available market ask, which is below `70000`).

TC-62 — Above-Market Order is NOT in Open Orders

Skipped if TC-29 failed.

Step 92: Open the **Open Orders** tab. Verify the above-market order is absent or shows as filled — it should not appear as a pending/cancellable entry since it filled immediately.

TC-63 — All Orders Shows Correct Details for Above-Market Order

Skipped if TC-29 failed.

Step 93: Open the **All Orders** tab. Find the row for the above-market order. Verify:

- Limit price field > 0 (approximately 70000)
 - Executed price > 0
 - Status shows "Done" (green color)
 - Side contains "Buy Full"
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TC-64 — Trade History Shows the Above-Market Buy Order

Skipped if TC-29 failed.

Step 94: Open the **Trade History** tab. Find the row matching the above-market order ID.

- If not found → log a suggestion ("order may still be processing") and skip assertions.

Step 95: If found, verify:

- Side contains "buy"
 - Executed price > 0
 - Amount > 0
 - Total > 0
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TC-65 — My Trades Shows Executed Market Price (Not Limit Price)

Skipped if TC-29 failed.

Step 96: Click the **My Trades** tab. Verify at least 1 entry exists. For the most recent entry verify:

- Executed price > 0
- Amount > 0

- Executed price is **less than** the limit price `70000` (the market filled at the actual ask, not the limit)
 - Executed price is within 1% of the confirmed executed price from the order dialog
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TC-66 — Balance After Above-Market Fill

Skipped if TC-29 failed.

Step 97: Navigate to Portfolio and capture a new balance snapshot. Compare against the pre-above-market snapshot (TC-60). Verify:

- USDT available balance decreased by approximately `executedPrice × amount` within 0.5% tolerance
 - BTC available balance increased by approximately `amount` within 0.5% tolerance
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TEARDOWN — After All Tests

Step 98: Close the browser.

Quick Reference Summary

Step Range	Area
1 – 6	Browser launch, login, dismiss dialogs
7	Navigate to Spot Trading page
8	Verify all page labels
9	Search for BTC/USDT currency pair
10 – 11	Mark pair as favorite, verify added
12 – 13	Unmark pair from favorites, verify removed
14	Select pair from ALL tab
15	Verify 24h ticker vs Binance
16	Verify order book column headers
17 – 19	Order book view switches (sell / buy / all)
20	Order book precision dropdown across all views
21 – 22	Order book LTP and ratio bar (suggestions)
23 – 24	Order book real rows and valid spread
25	Order book bid/ask vs Binance within 0.5%
26	Order book LTP vs Binance within 0.5%
27	Precision toggle restores LTP
28 – 29	Order book ask rows pre-fill buy form (5 rows)
30	Buy button label correct
31	Sell tab available BTC balance non-negative
32 – 35	25% / 50% / 75% / 100% percentage buttons
36	Pair header shows selected pair
37	Capture pre-order full balance snapshot
38 – 39	Enter limit price 57000 / total 8 USDT, verify estimated fee
40	Total field auto-calculates as price × amount
41	Fetch USDT available balance before order
42 – 46	Confirm buy order, detect fill, verify success

Step Range	Area
47 – 48	Balance snapshot after order (pending lock or immediate fill)
49	Market Trades panel headers
50	My Trades panel headers
51	Market Trades has real data rows
52	Market Trades prices vs Binance within 1%
53	Open Orders section visible with View All button
54 – 55	Open Orders row matches placed order
56	All Orders row matches placed order
57	USDT balance decreases by order total
58 – 59	Trade History tab validation
60 – 61	Trade History bottom widget validation
62 – 63	Cancel latest open order, verify balance restored
64 – 65	Full balance snapshot after cancel
66	Snapshot before multi-order Cancel All test
67 – 68	1st pending buy at 55000 USDT / 10 USDT total
69 – 70	2nd pending buy at 54000 USDT / 8 USDT total
71 – 72	3rd pending buy at 53000 USDT / 7 USDT total
73 – 74	4th pending buy at 52000 USDT / 9 USDT total
75 – 76	1st pending sell at 72000 USDT / 0.00011 BTC
77 – 78	2nd pending sell at 73000 USDT / 0.00012 BTC
79	Open Orders shows all 6 rows + Cancel All button
80	Balance shows USDT and BTC locked in inOrder
81	Trade History does not show pending orders
82	Cancel one order individually
83	5 orders remain after individual cancel
84	Cancel All removes all remaining orders
85	Open Orders shows "No data"

Step Range	Area
86	All Orders shows cancelled status (red)
87	Trade History does not show cancelled orders
88	Balance fully restored after Cancel All
89	Snapshot before above-market order
90 – 91	Place above-market buy at 70000 USDT / 7 USDT total
92	Above-market order not in Open Orders
93	All Orders shows Done (green) for above-market order
94 – 95	Trade History shows above-market buy
96	My Trades shows executed market price, not limit price
97	Balance after above-market fill: USDT down, BTC up
98	Close browser